Baishen Liang

Postdoctoral associate, Department of Neurology, Duke University
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EDUCATION

Doctor of Philosophy in Cognitive Neuroscience — **2017 – 2023**
University of Chinese Academy of Sciences, Beijing, China — *GPA: 3.94*

Exchanged Student, Department of Linguistics — **2016 – 2016**
University of California, Berkeley, Berkeley, CA, USA — *GPA: 3.78*

Bachelor of Science in Psychology — **2013–2017**
Central China Normal University, Wuhan, China — *GPA: 3.88*

WORK EXPERIENCE

Postdoctoral Associate — **July 2024 – present**
Department of Neurology, Duke University, Durham, NC, USA

- Lead a project to Design and establish an experiment to **study production-based verbal working memory maintenance and manipulation**. Collect, analyze, and maintain neurosurgical patients' intracranial electroencephalogram (iEEG) data.

Graduate Researcher — **September 2017 – June 2023**
Institute of Psychology, Chinese Academy of Sciences, Beijing, China

Technology Consultant for the Core Facility IPCAS (Nominated Position)
- Assisted in maintaining transcranial magnetic stimulation (TMS) facilities.
- Gave regular public lectures and courses for faculty and students.
- Provided one-on-one technical consultation for faculty and students requiring TMS.

Project 1: Bilateral Laryngeal Motor Cortex in Speech Perception in Noise
- Lead a project to apply functional magnetic resonance imaging (fMRI) to localize Mandarin speakers' laryngeal motor cortices. Designed and established a TMS experiment, and collected and analyzed data.

Project 2: Transcranial Alternative Electric Stimulation (tACS) on Auditory and Motor Cortex
- Led a project to establish a technical pipeline combining phase-controlled dual-site tACS and electroencephalogram (EEG) recordings for auditory cognitive neuroscience research.

Project 3: Cortical Functional Map for the Perception of Speech Features
- Systematically reviewed 148 published fMRI studies on speech perception and comprehension.

PEER REVIEWED PUBLICATIONS

1. **Liang, B.,** Li, Y., Zhao, W., Du, Y.* (2023). Bilateral human laryngeal motor cortex in perceptual decision of lexical tone and voicing of consonant. ***Nature Communications***, 14, 4710. [First author]

2. **Liang, B.** & Du, Y.* (2018). The functional neuroanatomy of lexical tone perception: an activation likelihood estimation meta-analysis. ***Frontiers in Neuroscience***, 12, 495. [First author]

3. **Liang, B.** & Du, Y.* (2017). Mechanisms of sensorimotor integration in speech perception [言语知觉中的感觉运动整合机制]. ***Science & Technology Review** [科技导报]*, 35(19):21-28. [First author]

MANUSCRIPTS UNDER REVIEW

1. Hu, W.#, **Liang, B.#**, Du, Y.* (2026). Singing enhances poetry learning and memorization in preschool children. ***Annals of the New York Academy of Sciences*** **(Revised and Resubmitted)**. [Co-first author]
2. Liu, J.#, Wang, Y.#, Feng, J.#, Zhao, H., He, Q., Yang, X.*, **Liang, B.* (Under review)** Successor representation supports structural learning of syllable sequences. [Co-corresponding author]
3. Wang, Y.#, Yu, K.#, Yin, S., **Liang, B.***, Wang, R.* **(Revised and Resubmitted)**. Theta and gamma transcranial alternating current stimulation (tACS) modulate Mandarin consonant and lexical tone perception. (Preprint available on ***bioRxiv***: 2025, 679546). [Co-corresponding author]

MANUSCRIPTS IN PREPARATION

1. **Liang, B.,** Earle-Richardson, A. M., Grant, G., Zafar, M., Frauscher, B., Southwell, D., Hickok, G., Cogan, G. B.* Sensory-motor mechanisms for verbal working memory. **In preparation.** [First author]

HONORS & AWARDS

- **Best Presentation Award in the 4th IEEE EMBS International Summer School of Neural Engineering. Tsinghua University**, Beijing, August 2018.
- **University of Chinese Academy of Sciences Student Grant (￥10,000)**. University of Chinese Academy of Sciences, Beijing. December 2017.
- **National Scholarship (￥8,000)**. Ministry of Education of the People's Republic of China, Beijing. November 2016.

PROFESSIONAL MEETINGS ATTENDED

- **Liang, B.,** Earle-Richardson, A. M., Grant, G., Zafar, M., Frauscher, B., Southwell, D., Hickok, G., Cogan, G. B.* Sensory-motor mechanisms for verbal working memory.
  - *Neuroscience 2025, Society for Neuroscience*. San Diego, CA, USA, November 2025 (poster).

- *Advances and Perspectives in Auditory Neuroscience*. San Diego, CA, USA, November 2025 (poster).
- *Society for the Neurobiology of Language 17th Annual Meeting*. Washington DC, USA, September 2025 (poster).
- *Duke Comprehensive Epilepsy Center Research Colloquium*. Durham, USA, September 2025 (slide presentation)
- *Duke Neurology TBS/DCEC Research Symposium*. Durham, USA, September 2025 (poster)
- *North Carolina Conference on Cognition 2025*. Chapel Hill, USA, April 2025 (poster).

▪ **Liang, B.**, Li, Y., Zhao, W., Du, Y.* Bilateral human laryngeal motor cortex in the perceptual decision of lexical tone and voicing of consonant (poster and presentation). *Society for the Neurobiology of Language Annual Meeting 2023*. Marseille, France, October 2023.

▪ **Liang, B.**, Hong, Y., Li, Y., Du, Y.* Causal involvement of auditory-motor low-frequency oscillation coupling in sentence perception under noisy conditions: a tACS study (poster). *Society for the Neurobiology of Language Annual Meeting 2023*. Marseille, France, October 2023.

▪ **Liang, B.**, Li, Y., Zhao, W., Du, Y.* A causal role of the human laryngeal motor cortex in lexical tone and voicing perception (poster and presentation). Neurobiology of Language: *Key Issues and Ways Forward II*. Nijmegen, Netherlands (online), March 2022.

▪ **Liang, B.**, Du, Y.* Neuroanatomy of lexical tone: an activation likelihood estimation meta-analysis (poster). *Organization for Human Brain Mapping Annual Meeting 2018*. Singapore, June 2018.

PEER REVIEW EXPERIENCE

***Nature*,** 2025

***Brain and Language*,** 2025

***International Journal of Multilingualism*,** 2025

Acta Psychologica Sinica [心理学报], 2024

TECHNICAL SKILLS

▪ **Programming languages**

- Matlab, Python, R, and C++.

▪ **Psychology**

- Psychophysics experimental designs.
- Statistics: linear models, logistic models, and drift diffusion models (DDMs).

▪ **Linguistics**

- Acoustic processing and feature extractions by Praat/Matlab.

▪ **Neuroscience**

- Electroencephalography (EEG), Magnetoencephalography (MEG), and intracranial EEG experimental design, data collection, preprocessing, and analyses.

- Transcranial magnetic stimulation (TMS) and transcranial electrical stimulation (tES) experimental design, data collection, and analyses.